

RESEARCH ARTICLE

Application of ELM Algorithm Incorporating AE Principles in Wireless Communication Signal Detection

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ABSTRACT With the development of society, the requirements for communication are getting higher and higher, but the detection performance of wireless communication signals is insufficient. To solve this problem, this paper combines the principle of Auto-encoder (AE) and the feature classification function of Extreme Learning Machine (ELM) algorithm to obtain AE-ELM algorithm, and builds a new wireless communication signal detection model based on this algorithm, hoping to improve the detection performance of wireless communication signals. The performance comparison experiment of the proposed AE-ELM algorithm shows that the bit error rate and average relative time complexity of the algorithm are 0.0004 and 0.5292s respectively, which is better than the comparison algorithm. Additionally, the effectiveness of the fusion algorithm-based signal detection model for wireless communication is investigated, and the findings demonstrate that this model outperforms the others with the lowest detection error rate of 0.0031. The above results demonstrate the high application potential of the novel wireless communication signal detection model and its ability to significantly improve the performance of wireless communication signal detection. Its application to wireless communication signal detection can greatly promote the development of the signal detection field in China.

INDEX TERMS AE, ELM, wireless communication, signal detection, deep learning.

I. INTRODUCTION

In modern society, communication technology has widely penetrated into all areas of social life, playing an extremely important role in promoting global economic development and information exchange and dissemination [1]. Signal detection is one of the most important problems in wireless communication system, but it is faced with some problems and challenges. First of all, with the popularity of wireless networks and the shortage of spectrum resources, signal interference and spectrum competition are increasingly serious, resulting in the decline of the accuracy of signal detection.

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Secondly, the frequency bands and protocols of different wireless communication technologies are diversified, which makes signal detection equipment need to have the compatibility and adaptability of a variety of technologies. In addition, the propagation path of wireless signals is complex and changeable, such as multipath effect, attenuation and blocking, etc., which also brings difficulties to signal detection. Therefore, it is an urgent problem to improve the accuracy and stability of wireless communication signal detection technology. Multiple-input Multiple-output (MIMO) technology is a major technology used to boost wireless communication performance and has drawn significant attention from academics [2]. Through the use of massive antennas and RF chains, this technology can improve wireless

communication performance by increasing the spectrum efficiency of communication signals, network capacity, and data transmission rate [3]. However, the multipath effect occurs in MIMO systems as the number of antennas increases, and the significant amount of interference information makes it more challenging to detect wireless communication signals. Therefore, finding a solution that can effectively suppress the multipath effect and improve the signal detection performance of MIMO systems has become an urgent problem in the field of wireless communication. Traditional MIMO systems generally use linear and non-linear algorithms as information detection algorithms, but these algorithms are unable to meet the high requirements of detection algorithms in terms of complexity and BER performance [4]. Deep learning networks, as intelligent learning algorithms that have been widely used in recent years, have good feature extraction and generalisation expression capabilities and are now successfully used in computer image recognition, language processing and other fields [5]. Among them, Auto-encoder (AE) network has the ability to reproduce the original input data, and Extreme Learning Machine (ELM) has the ability of high precision and fast operation [6]. For the purpose to create an improved wireless communication signal detection model, the AE-ELM algorithm—which is created by combining the AE network and ELM algorithm—is therefore combined. The contribution of this research is to propose a new signal detection method that can improve the performance and reliability of signal detection in wireless communication systems by integrating the AE principle and the ELM algorithm. It has important theoretical and practical significance to solve the problem of signal detection in the current wireless communication system. This research is innovative in the field of wireless communication. First of all, the innovation of ELM algorithm integrating the principle of autoencoder (AE) lies in introducing deep learning technology to the detection of wireless communication signals. The traditional signal detection algorithms are mostly traditional machine learning algorithms based on mathematical models and statistical methods, while the ELM algorithm integrating AE principle can learn higher-level abstract features from a large number of wireless communication signals through the feature extraction capability of the automatic encoder, thus improving the accuracy and robustness of signal detection. Secondly, the ELM algorithm integrating AE principle has a faster training speed in the training process, which is also an innovation point compared with traditional machine learning algorithms. Traditional machine learning algorithms often consume a lot of time and computing resources for training in the face of large-scale data sets, while ELM algorithm can complete training quickly in the process of one forward propagation and one back propagation by randomly initializing the output weight matrix, which greatly improves the training efficiency. In addition, ELM algorithm with AE principle is also innovative in feature extraction ability. As an unsupervised learning method, the autoencoder can learn the potential representation of the data from the unlabeled data, so as to better capture the characteristic information of the signal.

By integrating AE principle, ELM algorithm can better adapt to the feature extraction requirements of different signal types, and has wider applicability in wireless communication signal detection. To sum up, the title “Application Research of ELM algorithm integrating AE principle in wireless communication signal detection” is innovative in the field of wireless communication. By introducing deep learning technology and fast-training ELM algorithm, the accuracy, robustness and efficiency of signal detection can be improved, bringing new possibilities for research and application in the field of wireless communication.

II. RELATED WORKS

With the increasing maturity of the deep learning field, various deep learning algorithms have emerged, among which the ELM algorithm is widely used in various fields because of its high accuracy and running speed. The improved model was found to be significantly more accurate than the original model, and the model was shown to be practically applicable to video quality monitoring systems [7]. To address the issues of large storage capacity, high void probability, and insufficient integrity of the guiding catalogues in stellar sensors, Zhu et al. proposed a large field-of-view stellar sensor model combining the spherical spiral method and ELM [8]. After an effectiveness study, it was found that the model could further reduce the storage capacity of the guide catalogues while improving image uniformity and integrity compared to the traditional model. To solve the problem that the structure of the metal skeleton material is vulnerable when performing selective CO₂ adsorption, Kondo's team proposed incorporating ELM-13 into the metal skeleton material to control the interaction between the material layers. A comparative analysis of this method with other similar methods showed that this method is more efficient and has a more stable performance in separating CO₂ [9]. To improve the effectiveness and stability of the intelligent traffic management system, Cui et al. proposed the fusion of gravitational algorithm and ELM to construct an improved ELM traffic flow prediction model, which was compared with other advanced models and found to have the highest prediction accuracy and sensitivity [10]. Sui et al. combined ELM and packaging components such as pressure and temperature sensors to construct an environmental logging system to address the difficulty of performing measurement work in extreme environments such as high temperatures and high corrosiveness, and the method was found to be of practical use in empirical studies for normal field testing in extreme environments [11]. Since photovoltaic output is intermittently affected by weather conditions, Abdillah et al. proposed a nuclear limit learning machine (K-ELM) method for solar irradiance prediction. The empirical analysis of this method showed that this method can reduce the procurement cost, obtain accurate photovoltaic output, and improve the quality and safety of photovoltaic output by [12].

With the development of communication technology, there are more and more methods applied in the communication signal detection model. In response to the problems

of large computation and complex testing problems in millimetre-wave communication signal localisation models, Grossi led a team to propose the construction of an adaptive signal monitoring and localisation model based on a generalised information criterion, to verify the effectiveness of the model, and found that the model has the advantages of excellent identification and localisation capability and strong anti-interference capability [13]. Wei et al. proposed to combine the interference anonymous precoder with signal-to-noise ratio to construct an improved interference resistant anonymous precoder for the problem of easy uncovering of anonymous pass identities in networks, and the method was empirically studied to have the advantage of pass stability while providing better protection of anonymous people's private information [14]. To improve the anti-interference capability of SPAD against dead time, Huang led the team to propose combining SPAD with optical wireless communication to construct SPAD photon counting detection system, and conducted a comparative study with the traditional detection system, and found that the system has stronger anti-interference capability and more significant BER and SNR performance advantages [15]. To address the lack of physical layer security in the terahertz band, Gao led a team to propose the fusion of deep neural networks and artificial noise signalling mechanisms to construct a DNN terahertz secure communication model. After validation, it was found that the model has significantly higher secrecy rate than the traditional method, and the model is also more resistant to secrecy interruptions [16]. An underwater optical wireless communication channel classification optimisation system was developed by fusing deep learning algorithms with a joint channel classification system to address the issue of underwater optical wireless communication rate and stability limitations caused by the complex and variable underwater environment. After an empirical study, the system was found to be more resistant to environmental disturbances and have superior performance compared to the traditional system [17]. Due to the non-serious interference between wireless ultraviolet communication system, to solve this problem, Zhao et al. proposed a joint estimation algorithm based on the Bayesian principle to obtain the channel response coefficient and communication signal, the validation of the performance of the algorithm, found that the algorithm robustness and error rate than other algorithms better [18].

In summary, the ELM algorithm not only has great advantages in terms of computational speed, but can also guarantee learning accuracy at high speed, so it has excellent prospects for application in many fields. Many methods have been used in wireless communication signal detection research. This study integrates the ELM algorithm with the AE principle to create a new wireless communication signal detection model based on the ELM algorithm, in order to further improve the computational efficiency and accuracy of the wireless communication signal detection model. It is expected that the computational speed and accuracy of wireless communication signal detection will be improved, which will contribute to the development of the wireless communication field.

III. CONSTRUCTION OF A WIRELESS COMMUNICATION SIGNAL DETECTION MODEL BASED ON AE-ELM ALGORITHM

Traditional signal detection algorithms are difficult to separate signals under multi-dimensional complex wireless signals. Using the AE algorithm, the classification idea of ELM algorithm is used to realize the signal detection reduction, build the AE-ELM signal detection algorithm. The specific process of constructing AE-ELM model in this study is as follows: First, Autoencoder (AE) is used to extract features and reduce dimension of input data to reduce data dimensions and noise. Next, the data after feature extraction is used as input, and the Extreme Learning Machine (ELM) is used for training. ELM is a fast training artificial neural network algorithm, which has high learning ability and low computational complexity. In the training process, ELM will randomly initialize the weights and biases from the input layer to the hidden layer, and adjust the weights of the output layer by regularization method. Finally, by adjusting the parameters and structure of the AE-ELM model, the performance of the model is continuously optimized to achieve accurate detection of wireless communication signals. Through this process, the AE-ELM model can be used to extract and classify wireless communication signals effectively, and improve the detection performance and reliability of wireless communication systems.

A. APPLICATION OF THE AE-ELM ALGORITHM TO DETECTION MODELS

In the field of wireless communication, signal detection is an important task, which is used to distinguish different signal types and determine the presence or absence of signals. Common wireless communication signals include pulse amplitude modulation (PAM), orthogonal frequency division multiplexing (OFDM), multiple input multiple output (MIMO) and so on. Traditional wireless communication signal detection methods are usually based on rule modeling and predefined feature extraction, but these methods have some limitations. First, they are less adaptable for complex signal scenarios because the characteristics of the signal may be difficult to define or extract. Secondly, the accuracy and robustness of these methods may be affected by factors such as signal-to-noise ratio and multipath fading. In order to overcome these problems, artificial intelligence (AI) technology has been widely used in wireless communication signal detection in recent years. Adaptive reinforcement learning (AE) algorithm and Extreme Learning Machine (ELM) algorithm are widely used in signal detection models.

AE network is an unsupervised learning algorithm, which is an extended product formed by stacking the hidden layers of the network, which can reproduce the original input neural network [19]. The AE network is also an improvement on the traditional neural network algorithm, which is able to automatically extract small samples and unlabelled features automatically, so the network can better improve the running speed of the current manual extraction of features. Three layers of neural networks—an input layer, an implicit layer, and

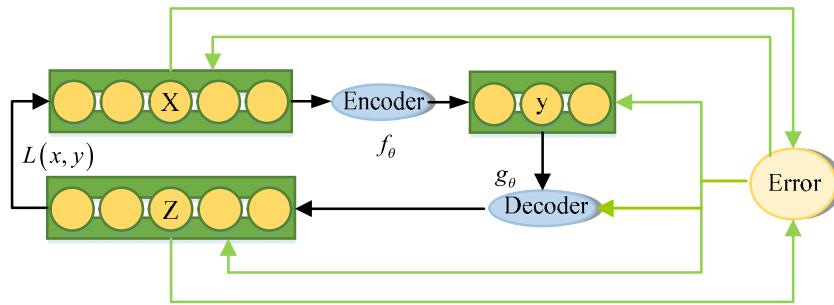


FIGURE 1. Auto-encoder network model.

an output layer—make up the majority of the AE network. Figure 1 shows the organisational structure of the network model. While the number of nodes in the input layer and the number of nodes in the output layer are kept constant, the number of nodes that make up the implicit, input and output layers can be changed as required.

As shown in Figure 1, the AE network can be divided into two parts: encoding and decoding. The coding structure in this instance runs from the input layer to the implicit layer, where the input data and intermediate data can be compressed using network information. The compressed information can be recovered using the decoding structure in Figure 1 that runs from the implicit layer to the output layer. In the encoding stage, the input vector x is mapped onto the intermediate layer representation y using the excitation function f_{θ} . In the decoding stage, the intermediate layer representation y is mapped using function g_{θ} to obtain the reconstructed data z . The encoding equation is shown in equation (1).

$$y = f_{\theta}(x) = s(wx + b) \quad (1)$$

In equation (1), b is the coding bias; w is the coding weight. The decoding equation is shown in equation (2).

$$z = g_{\theta}(y) = s(w'x + b') \quad (2)$$

In equation (2), b' is the bias term, w' is the decoding weight, w' is the transpose of w , and s is the sigmoid activation function. The AE model then uses unsupervised learning and training to reduce the reconstruction error, which is defined as the squared error between the input and output, in order to get stable network parameters. At this stage, the output value of the implicit layer differs from the sample of original data and can in some ways be used to characterize the original data. At this point, the cost function equation is shown in equation (3).

$$\begin{aligned} \min_{\theta} J_{AE}(\theta) \\ = \min_{\theta} \sum_{i=1}^n L(x_i, z_i) = \min_{\theta} \sum_{i=1}^n L(x_i, g_{\theta}(f_{\theta}(x_i))) \end{aligned} \quad (3)$$

In equation (3), θ is the adjustment parameter; x_i and z_i are the i th dimension input sample and output sample data respectively; n is the amount of input sample. The parameters to be adjusted are usually obtained by the classical gradient

descent method, as shown in equation (4).

$$\theta = \{w, b, w', b'\} \quad (4)$$

In equation (4), b and b' are the biases for encoding and decoding, and w and w' are the weights for encoding and decoding. The reconstruction error equation between the input and output data is shown in equation (5).

$$L(x, y) = \sum_{i=1}^n \|x_i - z_i\|^2 \quad (5)$$

In equation (5), x_i is the i th dimensional input sample data; z_i is the i th dimensional output sample data; and n is the amount of input samples. A good method for feature extraction is the AE algorithm with a non-linear transformation created by weights and biases. When the implicit layer and input-output layer have the same number of nodes, the extracted features can create an equidimensional representation of the input data. By altering the strength of the connections between neurons, the ELM, a single hidden layer feed forward neural network with quick learning capabilities, may approximate the system to achieve complicated nonlinear mapping. ELM are faster to train than traditional single hidden layer neural networks and maintain a higher learning accuracy. The ELM is a modified structure that builds on the classic single hidden layer neural network as a further advancement. Similar to the conventional single-hidden layer neural network, the ELM is fully connected between the input and hidden layers as well as between the hidden and output layer neurons. However, the connection weights of the ELM network's input layer are generated at random rather than having to be manually adjusted, which significantly reduces the amount of time required for computation. In Figure 2, the network architecture is displayed.

As shown in Figure 2, $a_{i,j}$ denotes the random connection weights between the i th input neural node and the j th hidden layer neural node, and b_j is the bias of the j th neural node, setting the training equation set as shown in equation (6).

$$\left\{x_i, t_i \mid x_i \in R^d, t_i \in R^m, i = 1, \dots, N\right\} \quad (6)$$

In equation (6), x_i and h_i are the training data vector and the output label data respectively. The output matrix equation

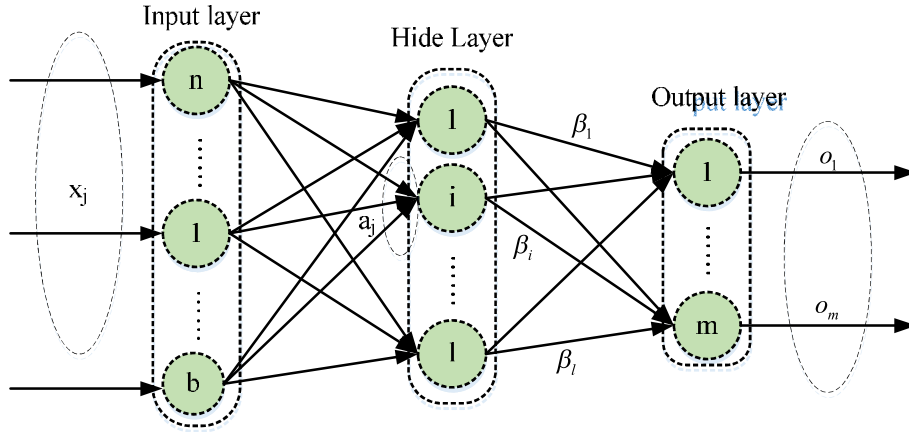


FIGURE 2. ELM network structure.

of the hidden layer is shown in equation (7).

$$H = \begin{bmatrix} h(x_1) \\ \vdots \\ h(x_N) \end{bmatrix} = \begin{bmatrix} h_1(x_1) & \cdots & h_1(x_N) \\ \vdots & \ddots & \vdots \\ h_L(x_1) & \cdots & h_L(x_N) \end{bmatrix} \quad (7)$$

In equation (7), L is the number of neurons in the hidden layer. The output value of the j th hidden layer node for the training sample x_n is shown in equation (8).

$$h_j(x_n) = g\left(\sum_i x_n(i) \cdot a_{ij} + b_j\right) \quad (8)$$

In equation (8), g is the output layer connection weight of the hidden layer activation function. Ridge regression theory is a complementary method to least squares, which reduces the probability of a “sick matrix” by increasing the activation factor, resulting in more stable results and better generalisation. The specific equation for ridge regression theory is shown in equation (9).

$$\beta = T \cdot H^T \left(\frac{1}{C} + HH^T \right)^{-1} \quad (9)$$

In equation (9), C denotes the activation factor term. the corresponding output function of the ELM network is shown in equation (10).

$$f(x) = T \cdot H^T \left(\frac{1}{C} + HH^T \right)^{-1} \cdot g(Wx + b) \quad (10)$$

In equation (10), HH^T denotes the singular matrix; T is the target output vector; and H^T is the implied layer output. Since the ELM algorithm can not only have fast training speed in the classification regressor, but also the deep network composed of its variant has the learning efficiency of the original ELM, the AE algorithm based on ELM is studied and the AE-ELM detection algorithm is constructed, as shown in Figure 3.

The process of the algorithm is divided into two phases, training and detection, as shown in Figure 3. In order to accurately classify and identify different categories of feature data, the ELM must first be trained using the labelled data

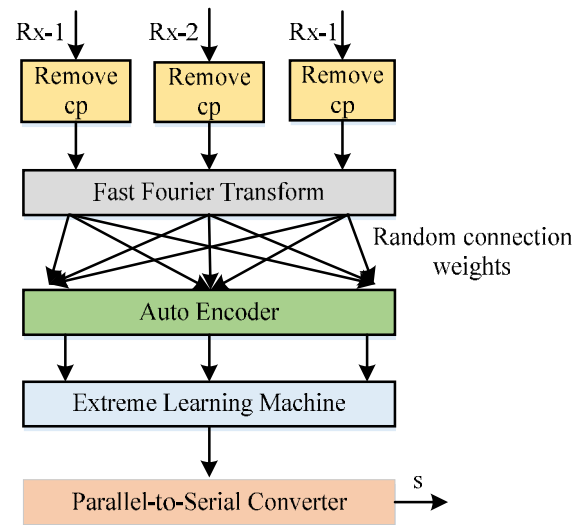


FIGURE 3. Block diagram of AE-ELM detection algorithm.

and the output feature data. This is done during the training phase, which mainly focuses on the automatic acquisition of the expression features of the input data and the new feature extraction capability of the sample data by the AE algorithm. This step requires initialising the AE and connecting the weights. The encoding equation is shown in equation (11).

$$h = f_{\theta}(W \cdot x + b) \quad (11)$$

In equation (11), W and b are the connection weights and bias terms at the coding end, respectively. The decoding equation is shown in equation (12).

$$x' = f_{\theta'}(W' \cdot h + b') \quad (12)$$

In equation (12), W' and b' are the connection weights and bias terms at the decoding end, respectively. The surrogate function needs to be regularised, and the specific equation for regularisation is shown in equation (13).

$$J = \sum \|x - x'\|^2 + \lambda \cdot \sum (W^2 + W'^2) \quad (13)$$

In equation (13), λ denotes the regularisation factor. The gradient descent optimisation algorithm is then used to optimise the minimisation cost function and bring the AE algorithm to a converged state, retaining the coding layer connection weights and bias terms, and using the output features as input to train the next AE. In addition, the appropriate number of AE layers is selected according to the actual situation, and the above training operation is repeated until all layers are trained. After the AE feature extraction training is completed, ELM classification training is carried out and the final output feature mapping data of the AE is used as the input of ELM, and the corresponding sent data labels are used as the target output to obtain the training parameters. The equation is shown in equation (14).

$$\beta = y \cdot (a \cdot h + b)^{-1} \quad (14)$$

h in equation (14) is the output label data. To sum up, the combination of AE and ELM algorithm can give full play to their respective advantages. AE can adjust ELM parameters adaptively to different signal scenes. ELM can use the features extracted by AE to quickly classify and detect signals. Therefore, the application of AE-ELM algorithm in wireless communication signal detection model has important background and significance. It can improve the detection performance and reliability, adapt to different signal scenarios, and provide technical support for the optimization and improvement of wireless communication system.

B. CONSTRUCTION OF AN ELM SIGNAL DETECTION MODEL BASED ON AE PRINCIPLE

The study builds a communication signal detection model based on a large-scale MIMO system—a wireless communication system that employs multiple transmit and receive antennas to improve data transmission efficiency and reliability by transmitting signals to multiple channels in multiple spatial dimensions—in order to more accurately model communication signal detection. MIMO systems typically employ multiple digital signal processing techniques such as frequency diversity, spatial multiplexing and beamforming to improve system performance and capacity [20]. The first is its capacity to boost data transmission speeds, which may be accomplished by employing numerous transmit and receive antennas without increasing bandwidth and power consumption. The second is its ability to increase the capacity of the system, which can be increased without increasing bandwidth and power consumption by transmitting the signal on multiple channels in multiple spatial dimensions. This is followed by its ability to reduce the effects of propagation noise, which can be reduced by transmitting the signal on multiple channels in multiple spatial dimensions, thereby increasing the reliability of the system. Finally, it supports processes with many inputs and different outputs. By transmitting data on many channels in multiple spatial dimensions, multiple-input multiple-output (MIMO) systems can increase system performance and capacity without increasing bandwidth and power consumption [21]. The MIMO system model is shown in Figure 4.

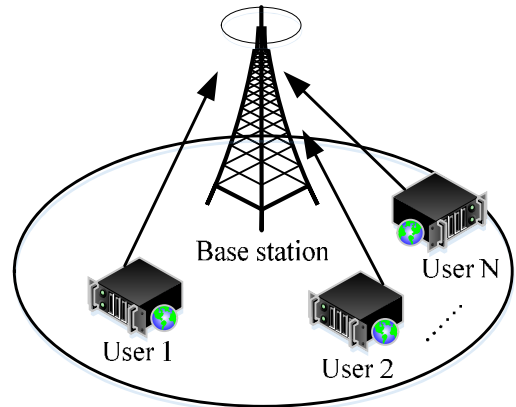


FIGURE 4. Multiple-in multipleout system mode.

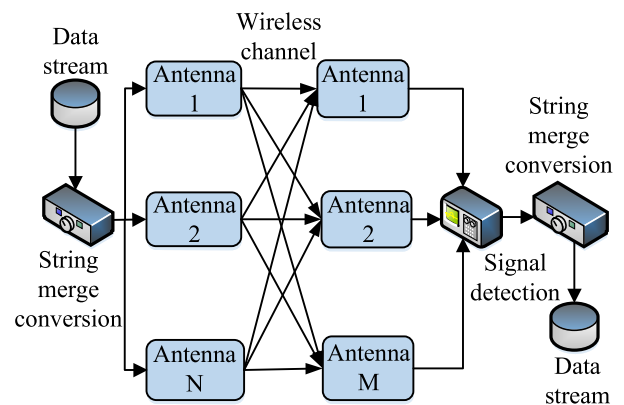


FIGURE 5. Signal detection system structur.

MIMO system detection technology is widely used in various fields because of its advantages of high efficiency, reliability and high capacity. However, the complexity of signal detection is great in classic MIMO systems because of the numerous antennas and RF chains, which has an impact on the accuracy of communication signal detection [22]. Therefore, it's critical to figure out how to increase the accuracy of its signal detection. Figure 5 illustrates the specific model structure of the MIMO system, which must be analyzed prior to the MIMO system being optimized.

As shown in Figure 5, any pair of sending and receiving antennas can access the wireless channel's channel characteristics, however numerous transmitting antennas can cause a multipath effect. Although MIMO systems make full use of the channel space due to multipath transmission, the multipath effect also makes it difficult to separate the signals by superimposing them at the receiver end. A simulation of the received signal state of a MIMO system with different signal-to-noise ratios has been carried out and the results are shown in Figure 6.

The horizontal and vertical axes in Figure 6 both represent the signal vector values and different shapes indicate different signals. Figure 6(a) shows the scatter plot of the received signals in the high SNR case. It can be seen from

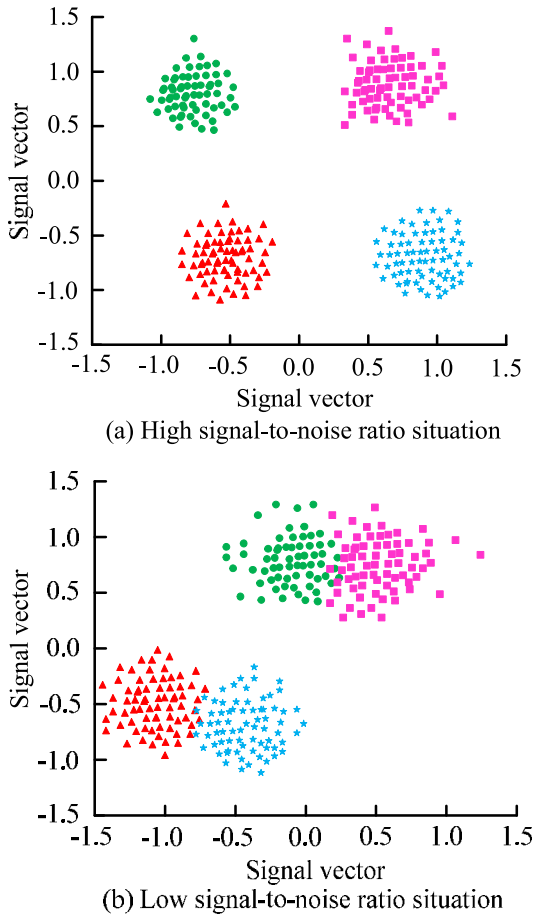


FIGURE 6. Simulation results of received signal states for MIMO system under different signal-to-noise ratio conditions.

Figure 6(a) that the received signals of different categories have clear boundaries in the high SNR state, and the signal characteristics of each category are more obvious. Figure 6(b) shows the scatter plot of the received signal in the low SNR case. From Figure 6(b), it can be seen that in the low SNR state, the sense of boundaries between different categories of received signals is not obvious, and the signals of each category are mixed together, and the category characteristics between each category are more blurred. From the above, it can be concluded that the poor propagation environment affects the quality of the received signals in the MIMO system, and it is this blurring of the boundaries between the signals that leads to the failure of the linear detection method. Therefore, the only way to separate the original information and improve the signal propagation environment is to collect the information from all the receiving antennas of the MIMO system. Due to dimensionality issues, the choice to reduce the difficulty of separation requires a certain degree of sacrifice of system information acceptance performance, and it is difficult to fit between computational complexity and detection performance. However, reception technology is the basis of communication signal detection, so it plays a crucial role in the performance of communication signal

detection models. Therefore, the research will improve on the MIMO wireless communication technology, aiming to reduce the signal separation difficulty of the signal detection technology without degrading the signal reception performance, and to obtain a relatively low-complexity detection algorithm model. The AE-ELM method is used to enhance the MIMO system in order to identify the communication signal detection algorithm model with the best performance and least amount of complexity. By using random mapping and quick training of the ELM algorithm to ensure that its input and output are consistent, the AE-ELM signal detection technique is suggested. Figure 7 depicts the AE-ELM signal detection algorithm model.

Figure 7 shows the structure of the ELM signal detection model incorporating AE principles. The model mainly consists of the AE-ELM structure training module, the feature mapping module and the ELM classification module. The channel conditions of the ELM signal detection system model with the AE principle are mobile and transmit and receive channels in free space. Unlike the AE algorithm where the connection layer weights are adjusted iteratively until the model converges, the retained parameters are used as parameters for the feature mapping model. The modified model generates weights and biases randomly through a Gaussian distribution, from which the input vector can be reconstructed. At this point the random encoding matrix h can be obtained through a random mapping approach using randomly obtained parameters calculated using the random mapping encoding equation shown in equation (15).

$$h = g(W \cdot x + b) \quad (15)$$

The mapping is more stable compared to the random parameters. The output weights are calculated as shown in equation (16).

$$a' = x \cdot h^T (I/C + h \cdot h^T)^{-1} \quad (16)$$

In equation (16), C denotes the activation factor term. In the feature mapping process, the connection parameters of the feature mapping layer are derived from the output layer weights, and the connection weight parameter of the feature mapping layer is set as a , which is calculated as shown in equation (17).

$$a = (a')^T \quad (17)$$

In equation (17), a' denotes the weight matrix from the output layer. The actual implied layer is more stable compared to the random features while maintaining the random feature characteristics. The specific calculation equation is shown in equation (18).

$$H = g(a \cdot x + b) \quad (18)$$

In equation (18), g is the implicit layer activation function; a is the connection weight; and b is the bias term. After the feature mapping module, it enters the ELM classification module. The structure of Figure 6 shows that the ELM classification module is trained using the corresponding label

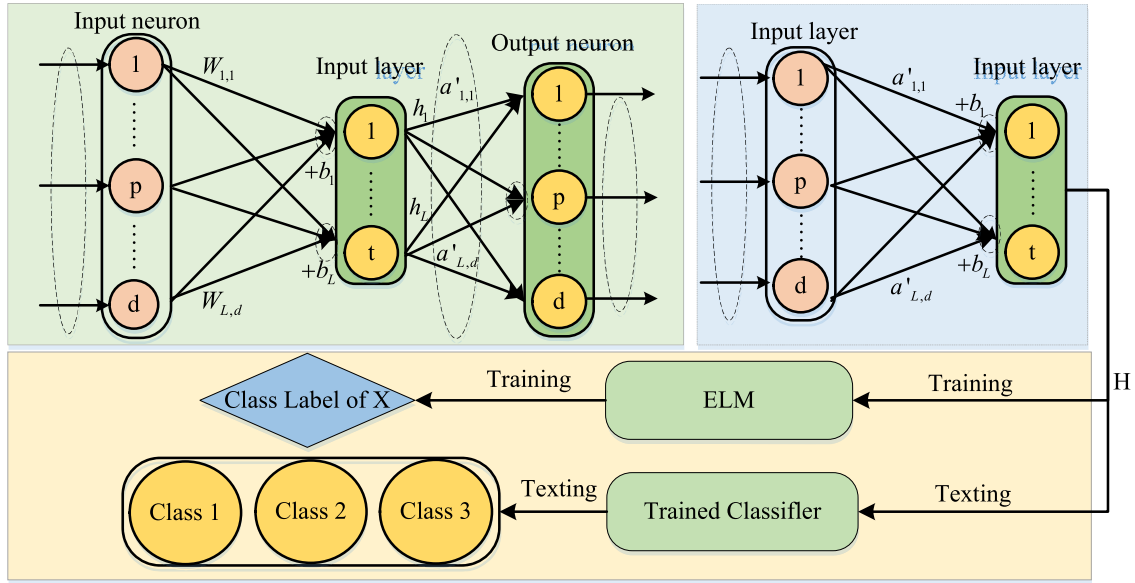


FIGURE 7. ELM signal detection algorithm model based on AE principle.

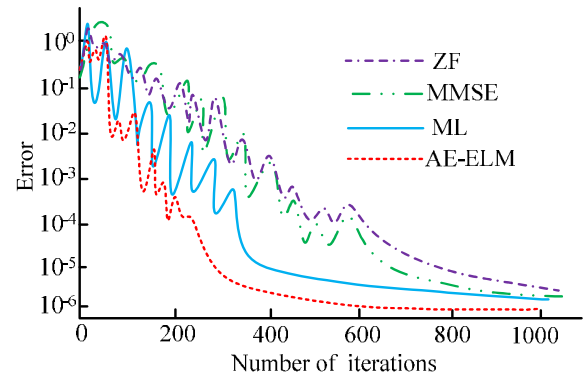
data after the improved model input data is mapped by the trained features. The training input sample is H , and the target output of the detector is the corresponding label data, which is the same as the input data x . In MIMO signal detection, the label data is used as the target signal while its corresponding signal type can also be selected according to the classifier features and reduced to the original signal as the target output according to the category information, thus achieving the final target information detection.

IV. COMPARATIVE PERFORMANCE ANALYSIS OF AE-ELM ALGORITHMS AND EMPIRICAL ANALYSIS OF SIGNAL DETECTION MODELS

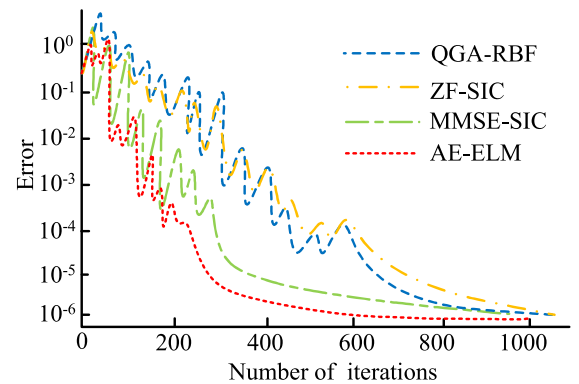
A. AE-ELM ALGORITHM PERFORMANCE ANALYSIS

The study evaluated the enhanced AE-ELM algorithm with other algorithms to compare the algorithm's performance with other algorithms in order to confirm the algorithm's great performance. The techniques used for comparison are Radial Basis Function (RBF), Maximum Likelihood Estimation (ML), Zero Forcing (ZF), Minimum Mean Square Error (MMSE), Successive Interference Cancellation (SIC), Quantum Genetic Algorithm (QGA), and Minimum Mean Square Error (MMSE). The study will compare the performance of the algorithms by setting the ratio of different signal noise power through MATLAB simulation experiments, using the training error, relative time complexity, detection signal BER (Bit Error Rate, BER) and other metrics. Figure 8 displays the training error outcomes of several algorithms used in the training process.

The results of the training error comparison for each of the seven distinct methods are plotted in Figure 8. Among them, Figure 8(a) shows the training error curves of the AE-ELM algorithm compared with the ZF algorithm, the MMSE algorithm and the ML algorithm. From Fig. 8(a),



(a) Comparison results of AE-ELM with ZF, MMSE, ML



(b) Comparison results of AE-ELM with QGA-RBF, ZF-SIC, MMSE-SIC

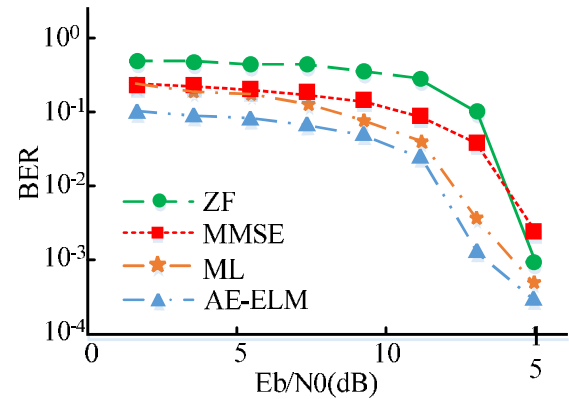
FIGURE 8. Comparison of training error results of different algorithms during the training process.

it can be seen that all four algorithms show fluctuations in the iterative process and an overall decreasing trend. The error curve of the AE-ELM algorithm is the first to stabilise

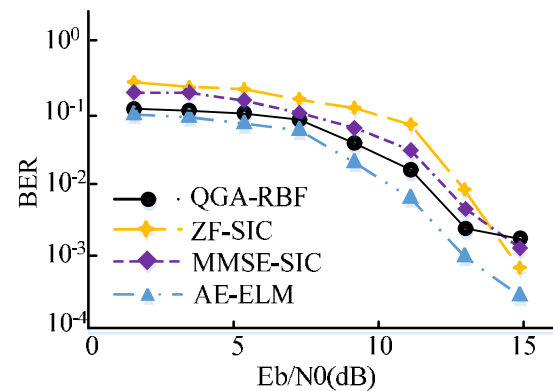
and its stabilised error is 0.000011, which is lower than the stabilised error of the other three algorithms. The performance of the algorithm is improved when the training error of the algorithm is smaller. Thus, the AE-ELM approach thus beats the ZF algorithm, the MMSE algorithm, and the ML algorithm in terms of the training error dimension, according to the results shown above. Figure 8(b) shows the training error curves of the AE-ELM algorithm compared with the QGA-RBF algorithm, the ZF-SIC algorithm and the MMSE-SIC algorithm. From Fig. 8(b), it can be seen that all four algorithms show fluctuations in the iterative process and an overall decreasing trend. Among the four algorithms, the error curve of AE-ELM is still the first to stabilize, and its stabilized error is 0.000010, which is also significantly lower than the stabilized error of QGA-RBF, ZF-SIC and MMSE-SIC algorithms. Therefore, it can be concluded from this result that the AE-ELM algorithm outperforms the QGA-RBF algorithm, ZF-SIC algorithm and MMSE-SIC algorithm in terms of the training error dimension. In summary, the AE-ELM algorithm proposed in the study has the best performance in terms of the training error dimension. Then, the performance of AE-ELM algorithm and six comparison algorithms are tested under different SNR conditions, and the test system is a 4×4 MIMO system simulated by computer. Model parameter settings in the experiment: the number of hidden layer nodes in the automatic coding machine was set to 2 to 4 times the length of the input vector, and AE used two different hidden layer activation functions in the experiment, a three-layer network was used in the experiment, and the length of the training data was set to 1600, and the test data was set to 10240. The BER results of the different algorithms are shown in Figure 9.

Figure 9 shows the BER performance curves of the AE-ELM algorithm and the comparison algorithm under different SNR conditions. From Figure 9(a), it can be seen that the BER of the ZF, MMSE, ML and AE-ELM algorithms all show a decreasing trend with increasing SNR. The BER of the AE-ELM algorithm is the highest at 1dB, 11.2%. At 15dB, the AE-ELM algorithm has the lowest BER of 0.04%, which is lower than the comparison algorithm. In the S/N range of 10-15dB, the BER performance decreases exponentially compared to the previous period. Figure 9(b) shows that the AE-ELM approach has a lower overall BER compared to the QGA-RBF, ZF-SIC, and MMSE-SIC detection algorithms. Taken together, these results indicate that the AE-ELM algorithm outperforms the comparator algorithms in terms of the BER dimension. Figure 10 shows the relative operation time complexity results for each algorithm in terms of ML for the case of 10 operation counts. The ZF and QGA-RBF curves were not included in the graphs for the study as the results were too far apart compared to the rest of the algorithms.

Figure 10 shows the results of each algorithm's variation in relative time complexity over the course of ten operations. As can be seen in Figure 10, the average relative time complexity of the AE-ELM algorithm varies within a narrow range of 0.5s, ranging from 0.5021-0.5491s, with an average relative



(a) Performance curve of traditional detection algorithm and improved algorithm detection algorithm

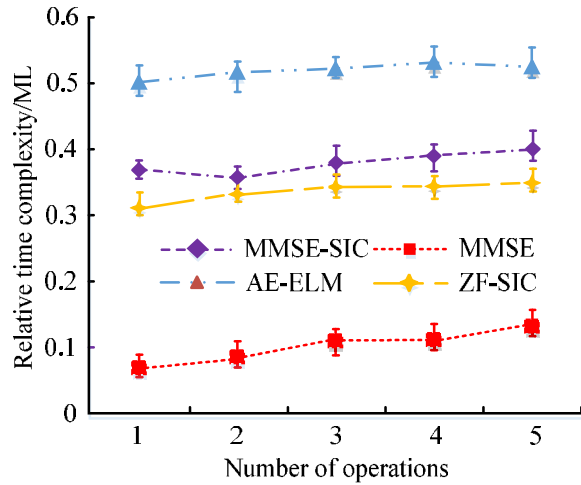


(b) Performance curves of popular and improved detection algorithms

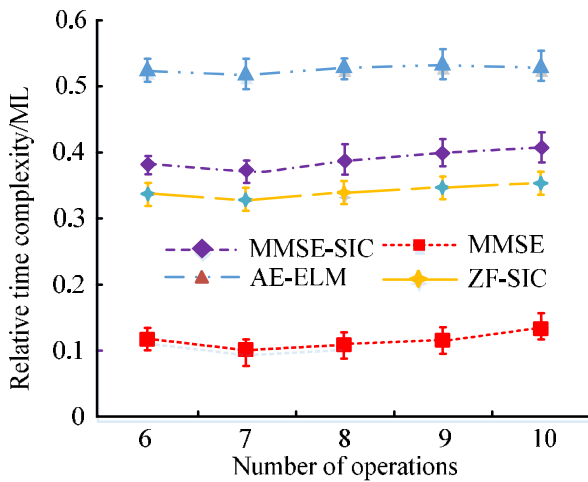
FIGURE 9. Detection performance curve of AE-ELM algorithm and other algorithms.

time complexity of 0.5292s. The average relative time complexity of MMSE-SIC is 0.3927 s. The average relative time complexity of ZF is 0.3119 s. The average relative time complexity of MMSE is 0.0482 s. The AE-ELM method performs better than the MMSE algorithm, the MMSE-SIC algorithm, the ZF-SIC algorithm and the ML algorithm in terms of time, according to the data presented above. Figure 11 compares the prediction accuracy of the four algorithms at different signal-to-noise ratio settings. The study also evaluated the performance of the AE-ELM algorithm against the MMSE-SIC method, the MMSE algorithm and the ZF-SIC algorithm.

In an environment with a signal-to-noise ratio of 1-20 dB, Figure 11 compares the prediction accuracy results of the AE-ELM algorithm with those of the MMSE-SIC algorithm, the MMSE algorithm and the ZF-SIC algorithm. As the signal-to-noise ratio increases, the accuracy of all four algorithms increases, as can be seen in Figure 11. Compared to the 83.7% accuracy of the ZF-SIC method, the 83.4% accuracy of the MMSE-SIC algorithm and the 82.8% accuracy of the MMSE algorithm, the accuracy of the AE-ELM algorithm is 85.2% at a signal-to-noise ratio of 1dB. The AE-ELM algorithm had the maximum accuracy of 95.3% at



(a) The relative temporal complexity of each algorithm during 1-5 operations



(b) The relative temporal complexity of each algorithm during 6-10 operations

FIGURE 10. The relative time complexity of each detection algorithm.

a signal-to-noise ratio of 20 dB, outperforming the ZF-SIC algorithm, the MMSE-SIC algorithm, and the MMSE algorithm by a wide margin (92.4%, 92.6%, and 91.3%, respectively). The aforementioned results demonstrate that, in various signal-to-noise ratio situations, the prediction accuracy of the AE-ELM algorithm suggested in the study is superior than that of the MMSE-SIC method, MMSE algorithm, and ZF-SIC algorithm. Therefore, it can be concluded that the AE-ELM algorithm outperforms the three comparison algorithms in terms of the prediction accuracy dimension. Combining the above four dimensions of training error, BER, relative computing time and prediction accuracy, the AE-ELM algorithm proposed in the study has the best performance. Since it can effectively increase the detection accuracy of the signal detection model and thus boost the growth of the wireless signal detection field, the study applies it to the wireless communication signal detection model.

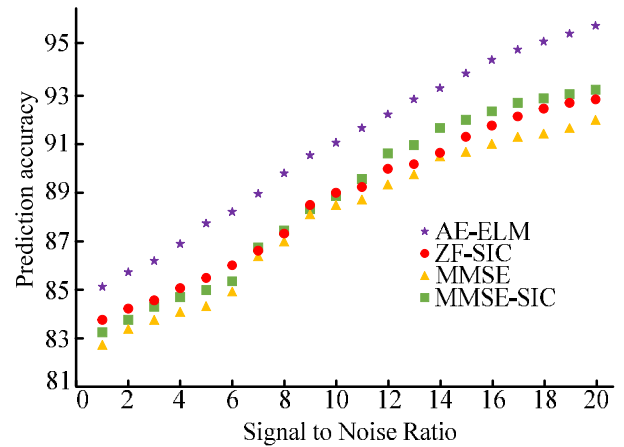
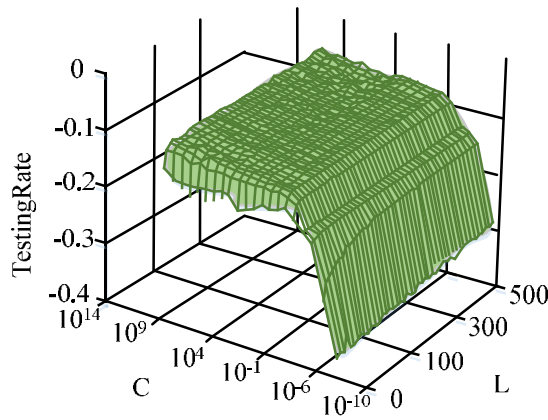


FIGURE 11. Comparison results of prediction accuracy of four algorithms.

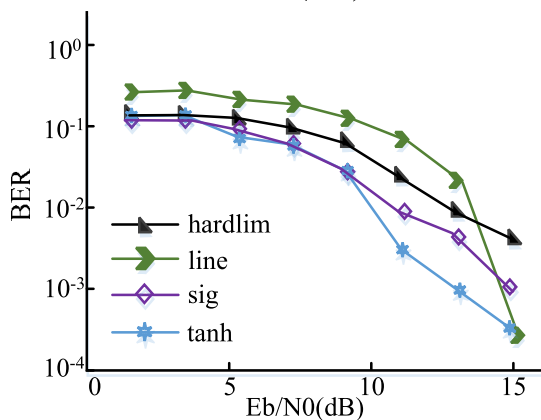
B. EMPIRICAL ANALYSIS OF AE-ELM SIGNAL DETECTION MODEL

After verifying the superiority of the AE-ELM algorithm, a new AE-ELM signal detection model is constructed by incorporating it into the wireless signal detection model. In order to conduct empirical analysis of the new AE-ELM signal detection model, the important parameters of the new AE-ELM signal detection model are first tested through simulation experiments, and the optimal parameters are selected to participate in the performance comparison experiments of the AE-ELM signal detection model. The activation factor C , the number of neural nodes in the hidden layer L , and the selection of the activation function are the three most important parameters in the novel AE-ELM signal detection model. Accordingly, performance tests on the activation factor C , the number of hidden layer neural nodes L , and different activation functions were performed in the study. The test results are displayed in Figure 12.

Figure 12(a) shows the BER difference of the AE-ELM detection model with the ML detection model as the standard for different parameters. The smaller the BER means the better its performance. Therefore the data at the highest point on the result curve should be selected as the best parameter for this model. From Fig. 12(a), it can be seen that the BER gap is small, i.e. the performance is optimal, for the activation factor C taken as 10-3-102 and L taken as 100-500. Since the optimal value of C is significantly affected by the signal-to-noise ratio, the study will set the parameter C to $1/\text{snr}$ and the parameter L to 120 in order to reduce the difficulty of model calculation. It has also been experimentally verified that the parameter values (C, L) can effectively reduce the BER of the AE-ELM signal detection model and can improve the detection performance of the model to a certain extent, with the rest of the environment and parameters remaining unchanged. The BER performance of the AE-ELM detection model for various activation functions is displayed in Figure 12(b). Figure 12(b) shows that while there is little difference in BER between the tangent function $\tanh$ and the sigmoid function when the signal-to-noise ratio



(a) Performance curves relative to ML under different (C, L) conditions



(b) Performance curves of popular and improved detection algorithms

FIGURE 12. Optimization test of important parameters in simulation experiments.

is 1-9dB, there is a significant difference between the two when the signal-to-noise ratio is 9-15dB. In summary, the tangent function tanh is the best, so tanh will be chosen as the activation function in subsequent experiments. Figure 13 displays the BER performance results for various size training sets for the new AE-ELM signal detection model.

Figure 13 illustrates how the BER performance of the new AE-ELM signal detection model improves with higher S/N ratio values and decreases when the S/N ratio is in the 6-13 dB range. The BER of the AE-ELM model decreases with increasing training set size, depending on the training set size. The detection model performs best when the training set size is between 1700 and 2700, resulting in a reduced BER. However, due to the limitation of the model's training set size, the training set size should not be too large to choose a good balance between performance and size. The above study verified the superiority of the AE-ELM algorithm and selected the optimal parameters and the optimal activation function. The study conducted performance comparison experiments with other comparable signal models and used the average detection time, BER and detection error rate as the comparison index to validate the practical application performance of

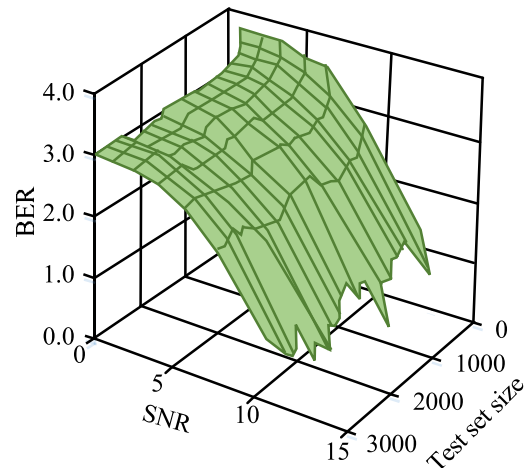


FIGURE 13. BER performance curves for different training set sizes.

the AE-ELM signal detection model. The results of average detection time and detection error rate for each signal detection model are shown in Figure 14.

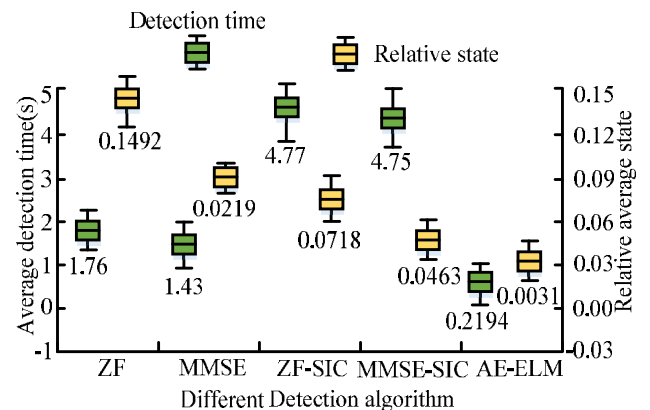


FIGURE 14. Average detection time and error rate of each detection algorithm.

The AE-ELM signal detection model has an average detection time of 0.219 seconds, which is the fastest of the algorithm models in terms of computational speed, as shown in Figure 14. The AE-ELM signal detection model also has the lowest average error rate of 0.0031, which is significantly lower than the other comparative signal detection models. According to the above results, the AE-ELM algorithm-based signal detection model is faster and more accurate than the ZF signal detection model, the MMSE signal detection model, the ZF-SIC signal detection model and the MMSE-SIC signal detection model. The results of the BER comparison of each signal detection model are shown in Figure 15.

Under various signal-to-noise ratio scenarios, Figure 15(a) displays the BER performance curves for the AE-ELM algorithm detection model and the traditional algorithm detection model. When the signal-to-noise ratio (SNR) is 1, the BER of the AE-ELM detection model is highest at 15%; when the SNR is 15, the BER is lowest at 0.03%. The

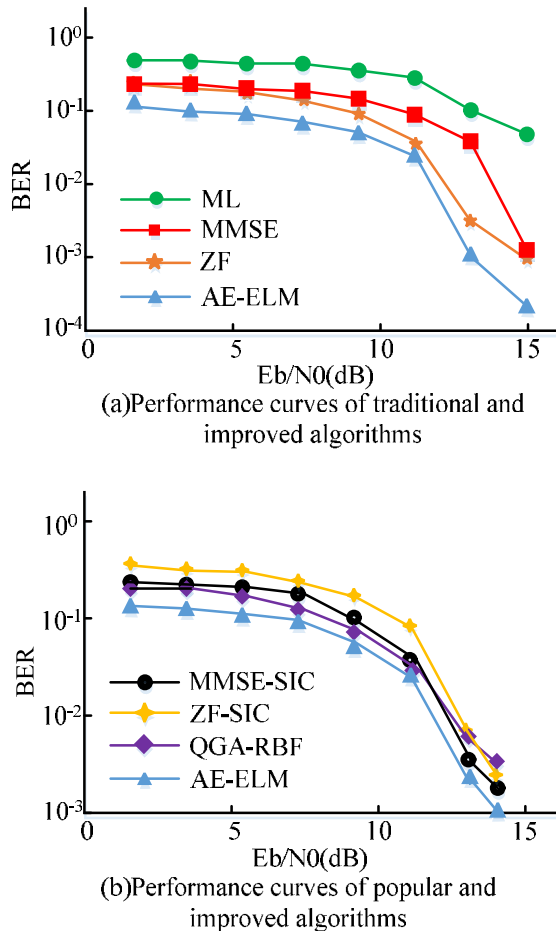


FIGURE 15. BER performance curve for each detection algorithm model.

BER performance curve for the common algorithm detection model and the AE-ELM algorithm detection model is shown in Figure 15(b). Figure 15(b) demonstrates that, at a signal-to-noise ratio of 1, the AE-ELM detection model has the maximum BER of 0.15%. The BER of the AE-ELM detection model is 0.1% at a signal-to-noise ratio of 14. The overall performance of the AE-ELM algorithm model is superior to that of the well-known model, however the BER curve is closer to that of the QGA-RBF algorithm detection model. When compared to the conventional and well-known algorithm detection models, the AE-ELM detection model performs best in terms of speed and BER. It also has the biggest benefit in terms of accuracy. The results of the empirical study demonstrate that the suggested AE-ELM algorithm model has promising futures in the area of wireless signal identification.

V. CONCLUSION

The detection accuracy of the classical wireless communication signal detection model cannot keep up with the current requirements due to the rapid development of communication technologies. A signal detection method including the AE principle and the ELM algorithm is proposed, and a

wireless communication signal detection model is built on it to improve the accuracy, speed and complexity of wireless communication signal detection. The performance of the AE-ELM algorithm proposed in the study is tested, and the results show that the AE-ELM algorithm has a BER of 0.0004 and its average relative time complexity is 0.5292s, both of which are better than the rest of the comparison algorithms. In addition, a performance comparison experiment was conducted on a modified wireless communication signal detection model based on the AE-ELM algorithm. The results showed that the AE-ELM signal detection model had an average detection time of 0.219s and an average error rate of 0.31%, both of which were significantly better than the comparative signal detection model. In addition, the AE-ELM detection model was found to have a BER of 0.03%, which is much lower than the other comparison models. The above results showed that the AE-ELM detection model outperformed the comparison model in terms of speed and accuracy, and had practical application value. The subsequent research direction of the experiment is to carry out empirical analysis on the performance of the algorithm under more complex signal environment. Looking forward to the future, ELM algorithm integrating AE principle will have broad application prospects in wireless communication signal detection. With the rapid development of wireless communication technology, the complexity and diversity of wireless signals are increasing, and the traditional signal detection algorithms are faced with many challenges. The ELM algorithm with AE principle has the advantages of fast training speed, good generalization ability and efficient feature extraction ability, which can effectively deal with these challenges. Firstly, the ELM algorithm based on AE principle can be applied to signal detection in wireless communication. There are a large number of signal types in wireless communication, such as modulation mode, transmission channel, interference and other different factors caused by signal changes. The ELM algorithm with AE principle extracts the feature representation of the signal through the automatic encoder, and classifies it by combining the ELM algorithm, which can better adapt to the detection requirements of different signal types and improve the accuracy and robustness of signal detection. Secondly, the ELM algorithm based on AE principle can be applied to the channel estimation task in wireless communication. The traditional channel estimation algorithm requires a lot of computing resources and time, and the assumptions of the channel model are strict. The ELM algorithm with AE principle can learn the channel feature representation through the automatic encoder, and combine the ELM algorithm to perform channel estimation, which can greatly reduce the computational complexity while ensuring the accuracy, and improve the efficiency and accuracy of channel estimation. In addition, the ELM algorithm with AE principle can also be applied to signal recognition tasks in wireless communication. In wireless communication, there are a lot of illegal signals, interference signals and so on, which is of great significance for the accurate identification of these signals. The ELM algorithm with AE principle can effectively distinguish various signals and improve the

accuracy and robustness of signal recognition by learning the feature representation of signals and combining the ELM algorithm for signal recognition. In short, ELM algorithm with AE principle has a wide application prospect in wireless communication signal detection. In the future, with the further development of wireless communication technology and the increasing application demand, the ELM algorithm based on AE principle will be continuously optimized and improved in technology, providing a more efficient and accurate solution for signal detection in the field of wireless communication.

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